

a PyBaMM-trained neural operator for second-life LFP RUL

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Second-life reuse of used lithium-iron-phosphate (LFP) cells for stationary energy storage is bottlenecked by characterisation cost: per-cell PyBaMM [1] calibration of degradation parameters against measured state-of-health (SoH) trajectories requires $\gtrsim 400$ cycles per cell [2]. Prior second-life work [3] and universal-differential-equation degradation [4] improve calibration *quality* without shrinking the cycling budget. We ask: *can a physics-conditioned neural operator, pretrained on PyBaMM-generated trajectories, predict the full SoH curve of a new cell from a single characterisation snapshot plus a short cycling window?*

We construct a two-stage pipeline. *Stage 1 — Physics corpus*: starting from PyBaMM’s Prada2013 LFP chemistry with per-cell BOL parameters identified from OCV/GITT/HPPC/RPT on a supplier-B fresh-cell cohort (OCV RMSE 6.8 mV median), we Sobol-sweep 300 simulations across five degradation channels (SEI kinetics, plating, LAM_{pos} , LAM_{neg}), producing 300 synthetic SoH trajectories with paired ground-truth θ . *Stage 2 — Neural operator*: a DeepONet [5] variant whose branch consumes a five-stream fingerprint (DCIR, RPT, first- K measured SoH, θ , protocol) and whose trunk consumes the target cycle number; output $\text{SoH}(n) = \text{SoH}_{\text{init}} - \text{softplus}(\langle b, t \rangle)$ is hard-monotonic by construction. θ -conditioning lets the operator absorb electrochemistry *directly* from Phase-1 param-ID rather than reconstructing it from characterisation.

Trained on the 300-sim corpus with $K = 50$ input cycles, the operator predicts held-out synthetic trajectories to **1.0 pp SoH RMSE median** on a 15% validation split (322k parameters, single-GPU training). Applied to a deep-second-life supplier-A cell at $K = 50$, the model forecasts RUL to a second-life EoSL threshold of $\text{SoH} = 0.40$: **1376 cycles under the measured operating protocol, ≈ 5000 cycles when projected under second-life BESS deployment at 0.25 C** via an empirical LFP C-rate fade-rate scaling ($3.5\times$ slowdown, Fig. 1). Both projections decouple the cycling budget from the extrapolation horizon.

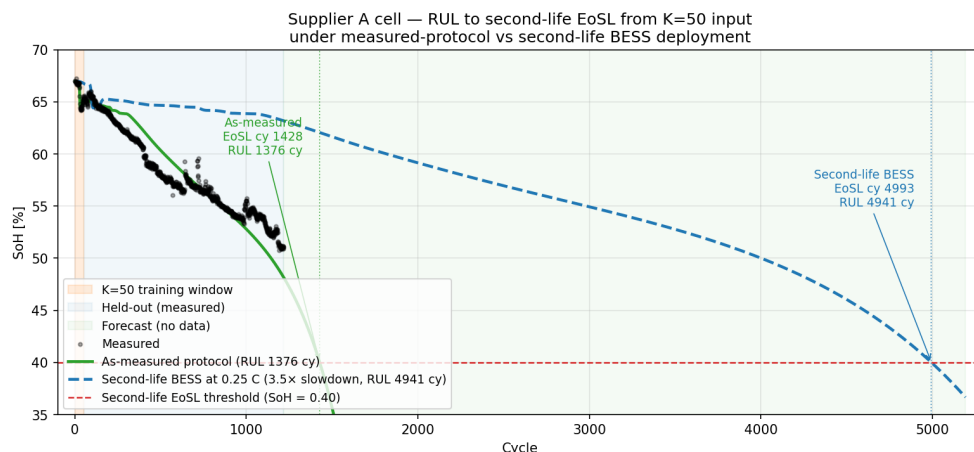


Figure 1: Neural-model RUL forecast for a deep-second-life supplier-A cell under two operating protocols. From $K = 50$ measured cycles (orange band), the green solid line is the as-measured-protocol extrapolation reaching EoSL at cycle 1428 (RUL 1376 cy); the blue dashed line projects the same cell under second-life BESS deployment at 0.25 C, reaching EoSL at cycle ~ 5000 (RUL 4941 cy). The neural-operator formulation retains the $K=50$ input budget and extends output to the full remaining-life horizon via PyBaMM-generated training trajectories.

References. [1] V. Sulzer et al., *PyBaMM*, J. Open Res. Softw. 9(1), 14–21 (2021). [2] M. Prada et al., *Simplified electrochemical and thermal model of LiFePO_4 -graphite Li-ion batteries*, J. Electrochem. Soc. 160(4), A616–A628 (2013). [3] O. Ouledboutaarija et al., *Advancing Second-Life Lithium-Ion Batteries*, PyBaMM Conf. 2025, p. 19. [4] J. A. Kuzhiyil et al., *Enhancing Generalisability of Physics-based Battery Degradation Using UDE*, PyBaMM Conf. 2025, p. 12. [5] L. Lu et al., *DeepONet: learning nonlinear operators for identifying differential equations*, Nat. Mach. Intell. 3, 218–229 (2021).